

COMPONENTS OF ELECTRONIC DEVICES

Filipovic D. Miomir

Components of electronic devices

This book is meant for those people who want to create electronic devices with their own hands. All components are illustrated and the circuit-symbol is explained in detail.

READ



BOOK DESCRIPTION

This book is meant for those people who want to create electronic devices with their own hands. All components are illustrated and the circuit-symbol is explained in detail. Both simple and complex examples are provided for the beginners. These include resistors, capacitors, transformers, transistors, integrated circuits, etc and each has its own symbol to represent it in an electrical or electronic diagram – called a circuit diagram. In order to understand how a certain device functions, it is necessary to know each symbol and the characteristics of the component. These are the things we will be covering in this book.

6

Like

2

+1

TABLE OF CONTENTS

Introduction	4.1 The working principle of a transistor	8.1 Microphones
	4.2 Basic characteristics of transistors	8.2 Speakers
1. Resistors	4.3 The safest way to test transistors	8.3 Headphones
1.1 Introduction to Resistors	4.4 TUN and TUP	8.4 Examples
1.1.1 Resistor Markings	4.5 Practical example	
1.2 Resistor Dissipation	5. Diodes	9. Opto-electronic components
1.3 Nonlinear resistors	5.1 Introduction to Diodes	9.1 Introduction to Opto-electronic components
1.4 Practical examples with resistors		9.1.1 Examples

- 1.5 Potentiometers
- 1.6 Practical examples with potentiometers

2. Capacitors

- 2. Introduction
- 2.1 Block-capacitors
 - 2.1.1 Marking the block-capacitors
- 2.2 Electrolytic capacitors
- 2.3 Variable capacitors
- 2.4 Practical examples

3. Coils and transformers

- 3.1 Coils
- 3.2 Transformers
- 3.3 Practical examples with coils and transformers

4. Transistors

- 4. Introduction to Transistors

- 5.1 Diode identification
- 5.2 Diode characteristics
- 5.3 Practical examples

6. Thyristors, triacs, diacs

- 6. Introduction to Thyristors, triacs, diacs
- 6.1 Practical examples

7. Integrated circuits

- 7. Introduction Integrated circuits
- 7.1 Analog integrated circuits
- 7.2 Digital integrated circuits
- 7.3 Practical examples

8. Microphones, speakers and headphones

- 8. Introduction to Microphones, speakers and headphones

10. Other components

- 10. Introduction to Other components
- 10.1 Relays

11. Checking Components

- 11. Introduction to Checking Components
- 11.1 Diodes and Transistors
- 11.2 Transformers and coils
- 11.3 Capacitors
- 11.4 Potentiometers
- 11.5 Speakers and headphones

12. Conductivity probe

- 12. Introduction to Conductivity probe
- 12.1 Semiconductors check
- 12.2 Checking other components

Book Name: Components of electronic devices

Author: Filipovic D. Miomir

Publication: January 1, 2003

Date:

Hashtag: # understandingelectronicscomponents

Keywords/Tags: components,electronic,devices,resistors,capacitors,t

Copyright: 2003 by MikroElektronika.

Components of electronic devices © MikroElektronika. All Rights Reserved, except where otherwise noted.